Configure CESM 2.2.2 to run on Derecho NCAR system (derecho.hpc.ucar.edu)

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NB: `Grayed terminal commands that appear on multiple lines must be entered as a single command line.`

1. Checkout release-cems2.2.2 branch:

```
git clone -b release-cesm2.2.2
https://github.com/ESCOMP/CESM.git cesm2.2.2
```

```
cd cesm2.2.2
```

```
export CESM_ROOT=$PWD
export CIMEROOT=$CESM_ROOT/cime
```

2. Get a higher version of CIME component version for compatibility with Derecho, branch *cime5.8.37*, instead of *cime5.8.32.9*. Edit the `Externals.cfg` with the text editor or streatmline "sed" editor:

```
sed -i 's/cime5\.8\.32\.9/cime5.8.37/g' Externals.cfg
```

> *Note: Another way to get a higher version of CIME without changing Externals.cfg:*
> *a) Proceed to the next step (3), checking out all the external subcomponents;*
> *b) Check out the required version of cime:*
>
> ```
> cd $CESM_ROOT/cime
> git checkout cime5.8.37
> ```

3. Checkout the CESM components:

```
./manage_externals/checkout_externals
```

4. Create a directory `$CESM_ROOT/cime_config/machines/` with files config_machines.xml and config_batch.xml, by copying it from a template:

```
cp -vr
/glade/work/nperlin/CESM2/cesm2.2.2/cime_config/machines_template
$CESM_ROOT/cime_config/machines
```

Edit `$CESM_ROOT/cime_config/machines/config_machines.xml` for Derecho, configure *CIME_OUTPUT_ROOT*, *DOUT_S_ROOT* variables as needed to reflect your tests and disk space.

5. To create a new case directory under </YOUR/CASE/DIR>, run the following script as shown below, specifying arguments for resolution, compset, project as needed (arguments with --res, --compset, --project); they may differ from the example shown.

```
cd $CESM_ROOT

$CIMEROOT/scripts/create_newcase --case </YOUR/CASE/DIR> --res
f09_g17 --compset B1850 --project UMIA0045 --machine derecho
--extra-machines-dir $CESM_ROOT/cime_config/machines 2>&1 | tee
log.create.newcase.001
```

A directory </YOUR/CASE/DIR> for the new case will be created.
It is practical to create a *newcase.sh* script for the record/repeating step 5:

```
vim /glade/work/nperlin/CESM2/cesm2.2.2/cime/scripts/newcase.sh
```

(paste; single long line:)

```
$CIMEROOT/scripts/create_newcase --case </YOUR/CASE/DIR> --res
f09_g17 --compset B1850 --project UMIA0045 --machine derecho
--extra-machines-dir $CESM_ROOT/cime_config/machines
```

```
chmod +x /glade/work/nperlin/CESM2/cesm2.2.2/cime/scripts/newcase.sh
```

6. Prepare to run a case setup (case.setup)

```
cd </YOUR/CASE/DIR>
export CASEROOT=$PWD
```

a) Adjust the RUNDIR if needed.
Query current settings for CASEROOT, RUNDIR, DOUT_S_ROOT variables:

```
./xmlquery CASEROOT RUNDIR DOUT_S_ROOT
```

If you need to change any of these variables, use `./xmlchange` as in the example below. CASEROOT may be staged in `/glade/work/${USER}`, but you may want to stage `${RUNDIR}` under `${SCRATCH}` for more efficient I/O. Create a directory as needed and reassign to RUNDIR (edit path or `<CESM_EXPT_DIR>/<CASE_DIR>` as your see fit):

```
mkdir -p ${SCRATCH}/<CESM_EXPT_DIR>/<CASE_DIR>/run

cd $CASEROOT
./xmlchange RUNDIR=${SCRATCH}/<CESM_EXPT_DIR>/<CASE_DIR>/run
```

Rerun query to verify the change:

./xmlquery RUNDIR

b) Change the default wave model setup to ask for fewer ntasks for WAV (`ntasks` to 36 seems to work well):

```
./xmlchange NTASKS_WAV=36
```

c) MARBL: Prevent the biogeochem model MARBL from saving hundreds of variables to the ocean output:

```
echo ' ' > SourceMods/src.pop/ecosys_diagnostics
```

d) If needed:

Add daily output of momentum fluxes from the ocean model, as from Sara Larson's work */glade/campaign/univ/uncs0054/slarson/add_daily_POPtau*

Daily wind stresses to be added to the ocean model output (TAUX_2, TAUY_2).
Under $CASEROOT, we now have a directory `${CASEROOT}/SourceMods` that contains modified code for the CESM2. This is where all the code changes go that override the default code

Place changes for TAUX_2 and TAUY_2 under

```
${CASEROOT}/SourceMods/src.pop/
     forcing.F90,
     gx1v7_tavg_contents
```

```
cd $CASEROOT
```

(E.g., copy from the adapted CESM2.2.2 code from:
/glade/work/nperlin/CESM2/EXPT/Atm10_Ocn1/SourceMods/src.pop)

```
cp -pv /glade/work/nperlin/CESM2/EXPT/Atm10_Ocn1/SourceMods/src.pop/*
.
${CASEROOT}/SourceMods/src.pop/.
```

e) Run the `case.setup` command:

```
cd $CASEROOT
./case.setup 2>&1 | tee log.case.setup.001
```

7. Build the case

a) Clean the build in case of an old build present:

```
./case.build --clean
```

b) Use default modules to build interactively. Use “module reset” to reset a list of modules to the default for the clean environment:

```
module reset
module list

Currently Loaded Modules:
  1) ncarenv/24.12 (S)   3) intel/2024.2.1        5)
libfabric/1.15.2.0   7) hdf5/1.12.3
  2) craype/2.7.31       4) ncarcompilers/1.0.0   6)
cray-mpich/8.1.29    8) netcdf/4.9.2
```

c) Verify the `nc-config` and `nf-config` commands are valid as set by the modulefiles, then set the NETCDF env variables:

```
nc-config --prefix
```

Prompt:
```
/glade/u/apps/derecho/24.12/spack/opt/spack/netcdf/4.9.2/packages/net
cdf-c/4.9.2/oneapi/2024.2.1/6ont
```

```
nf-config --prefix
```

Prompt:
```
/glade/u/apps/derecho/24.12/spack/opt/spack/netcdf/4.9.2/packages/net
cdf-fortran/4.6.1/oneapi/2024.2.1/fq5j
```

```
export NETCDF_C_PATH="$(nc-config --prefix)"
export NETCDF_FORTRAN_PATH="$(nf-config --prefix)"
export NETCDF_PATH="$(nc-config --prefix)"
```

d) Add the following to the bottom of the POP namelist file `user_nl_pop`:

```
n_tavg_streams = 3
```

e) Run the case.build command:

```
qcmd -A UMIA0045 -- ./case.build 2>&1 | tee log.case.build.002
```

8. Adjust configuration parameters and run the case:

a) Adjust runtime parameters:

```
./xmlquery EXEROOT
./xmlquery STOP_OPTION,STOP_N
./xmlchange STOP_OPTION=nmonths,STOP_N=6
./xmlquery RESUBMIT
./xmlchange RESUBMIT=1
./xmlquery DOUT_S_ROOT
./xmlchange DOUT_S=TRUE
./xmlquery JOB_WALLCLOCK_TIME
./xmlchange --subgroup case.run JOB_WALLCLOCK_TIME=02:00:00
./xmlchange --subgroup case.st_archive JOB_WALLCLOCK_TIME=00:20:00
```

b) Submit a case:

```
./case.submit >& log.case.submit.001 &
```